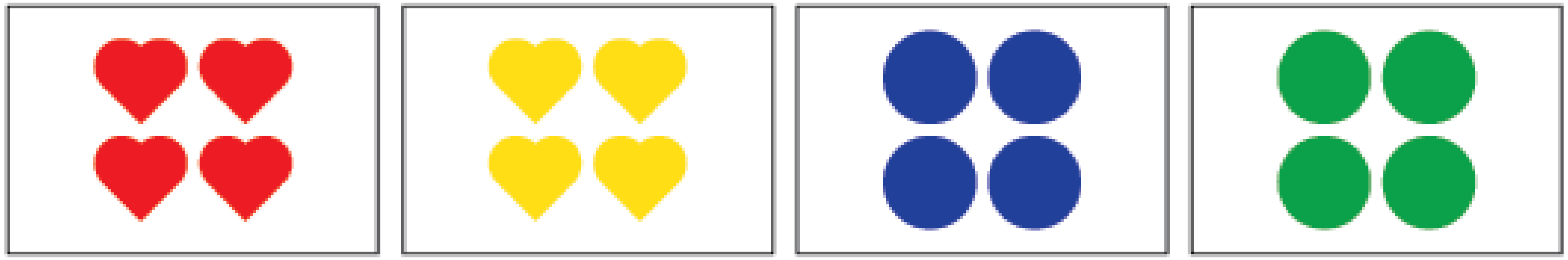

Quad Packing

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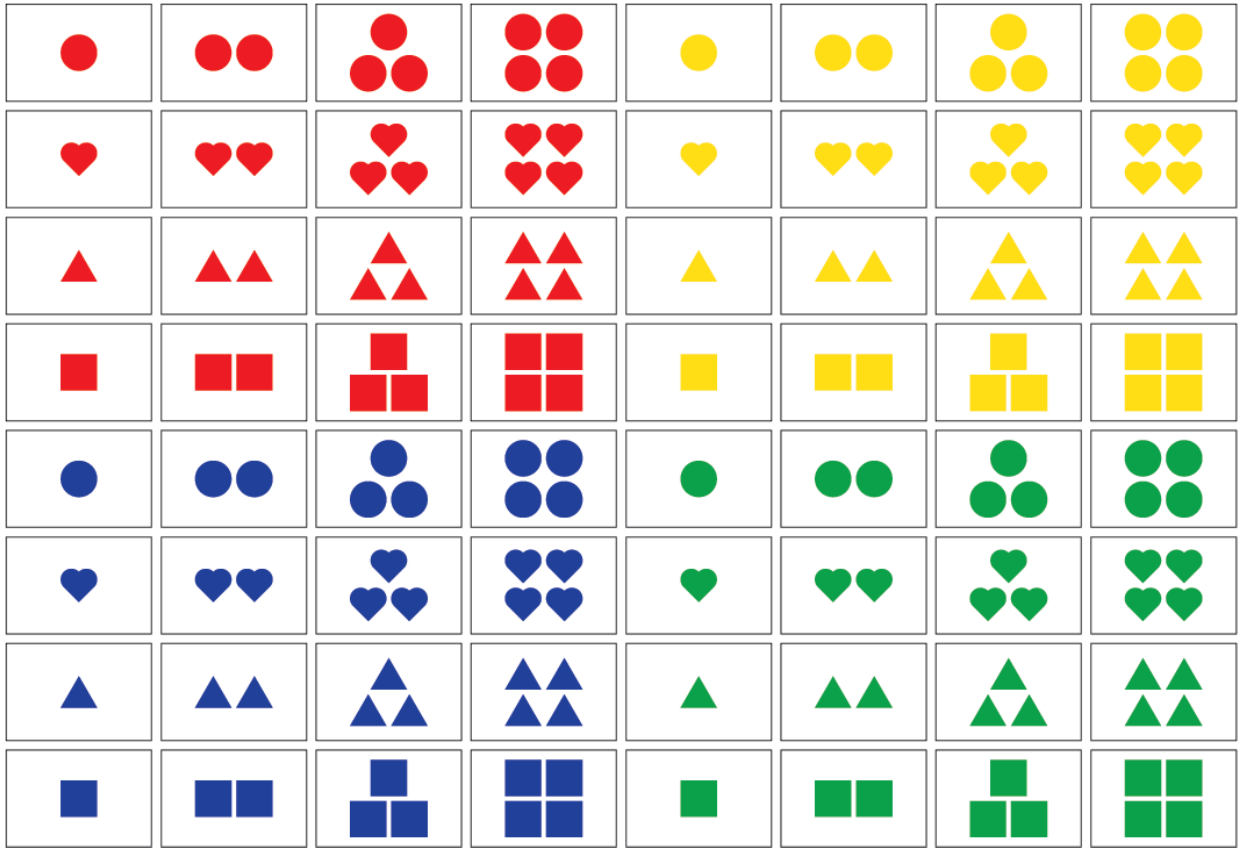
MARCH 1, 2025

Outline

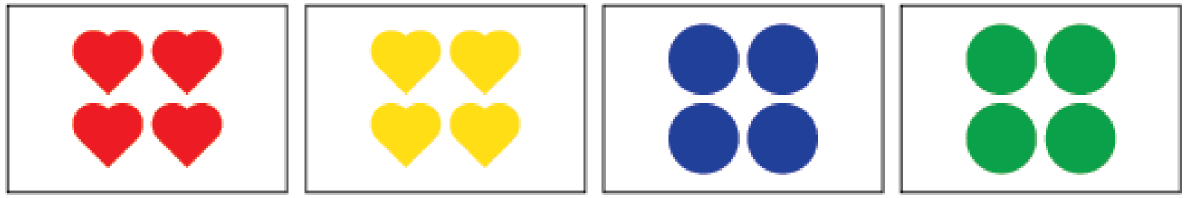
I. Fundamentals of Quads
II. The Problem
III. Results
IV. Code
V. Conjectures

What is Quads?



What makes a quad?

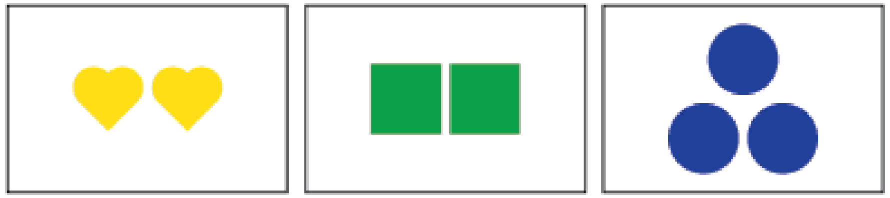
Expressions of attributes are all the same, all different, or in pairs.



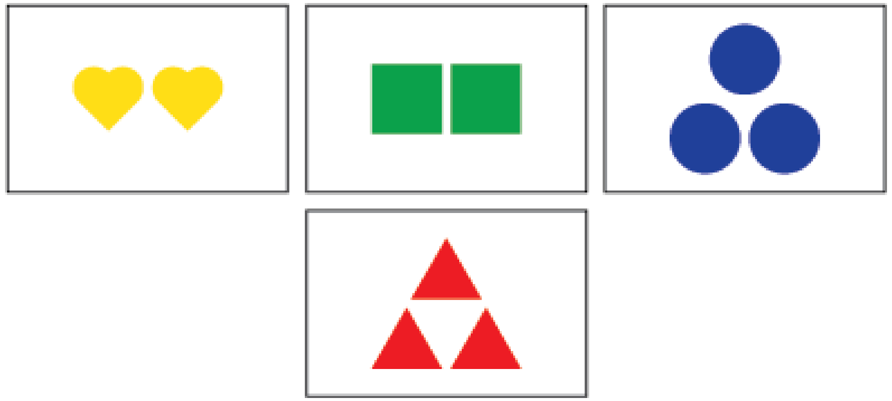
Number is all the same, shape is in pairs, and color is all different

Fundamental Theorem of Quads

For any three cards, there exists a unique fourth that completes a quad.



What card completes the quad?



Vector Representation & Geometry

We can assign vectors in $\mathbb{Z}_2^6$ to each card in Quads by using the following table.

$\mathbb{Z}_2^2$	number	color	shape
$(0,0)$	1	red	circle
$(0,1)$	2	yellow	heart
$(1,0)$	3	green	triangle
$(1,1)$	4	blue	square

Ex.  $=(0,0,1,0,1,1)$

Vector Representation & Geometry

Theorem: Distinct cards a, b, c, d are a quad iff $a + b + c + d = 0$.

Corollary: Given cards a, b, c, the card $a + b + c$ completes the quad.

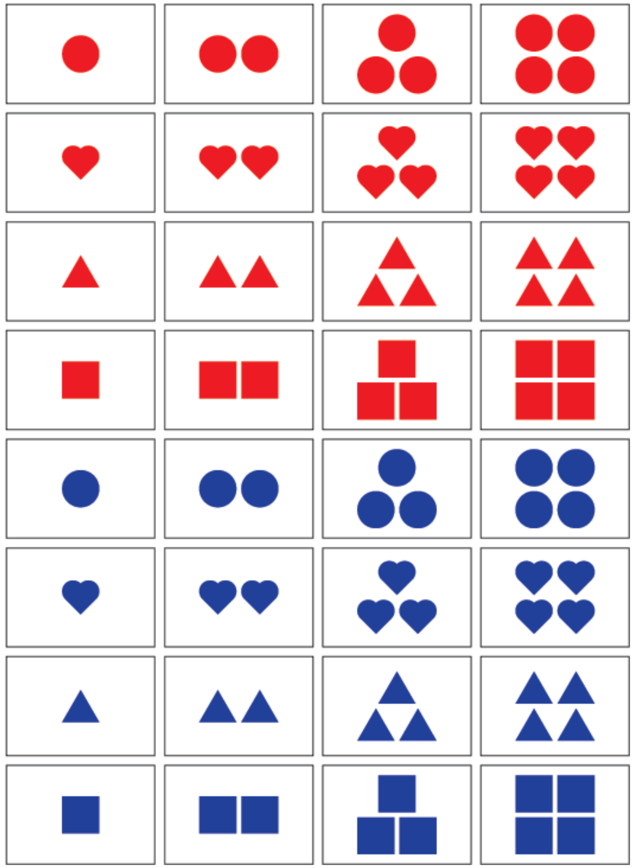
Corollary:

$$\begin{array}{r} a + b + c + d = 0 \\ a + b + e + f = 0 \\ \hline c + d + e + f = 0 \end{array}$$

Quad-2^d

Usually $d = 6$: three attributes with four expressions of each.

Previous theorems hold for any d.



The Problem

In Quad-64, what is the maximum number of quads in k cards?

Alternatively: What is the maximum number of planes that can be made with k points in $\mathbb{Z}_2^6$?

Let $q(k, d) =$ maximum number of quads given k cards in $\mathbb{Z}_2^d$.

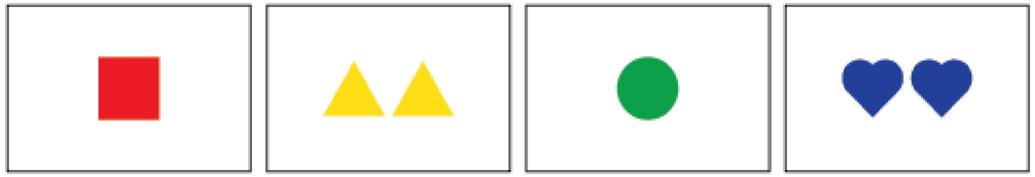
Small k

$q(4, d) = 1$

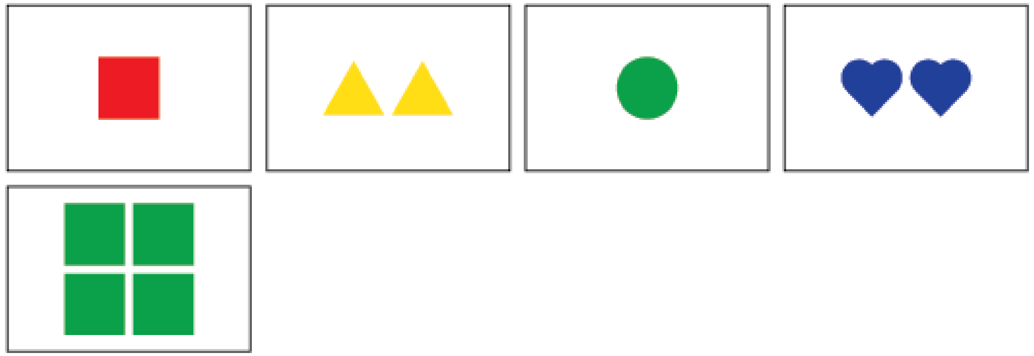
- *Trivial*

$q(5, d) = 1$

- *Proof:*
  - By FToQ, $q(5, d) = 1$



4 cards with 1 quad

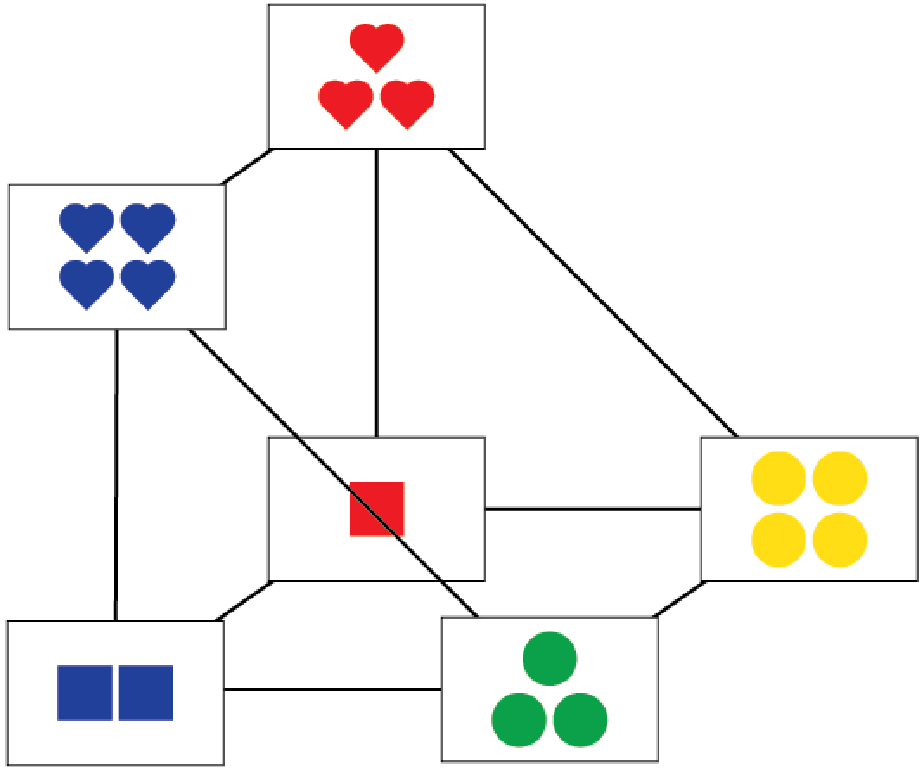


5 cards with 1 quad

Small k

$q(6, d) = 3$

- *Proof:*
  - Form two quads that share two cards, by the corollary this forces a third

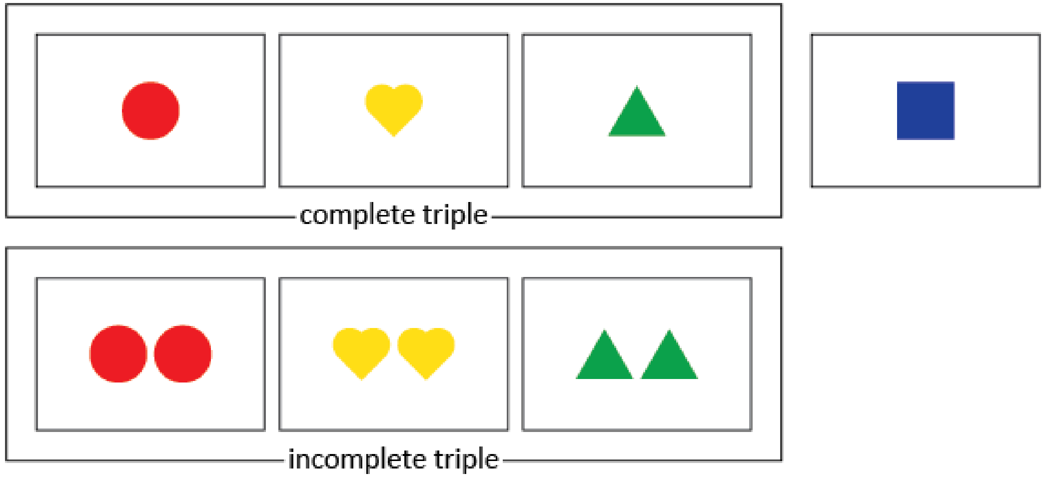


6 cards with 3 quads

An Upper Bound

Statement: For $4 \le k \le 2^d,\ q(k,d) \le \frac{\binom{k}{3}}{4}$.

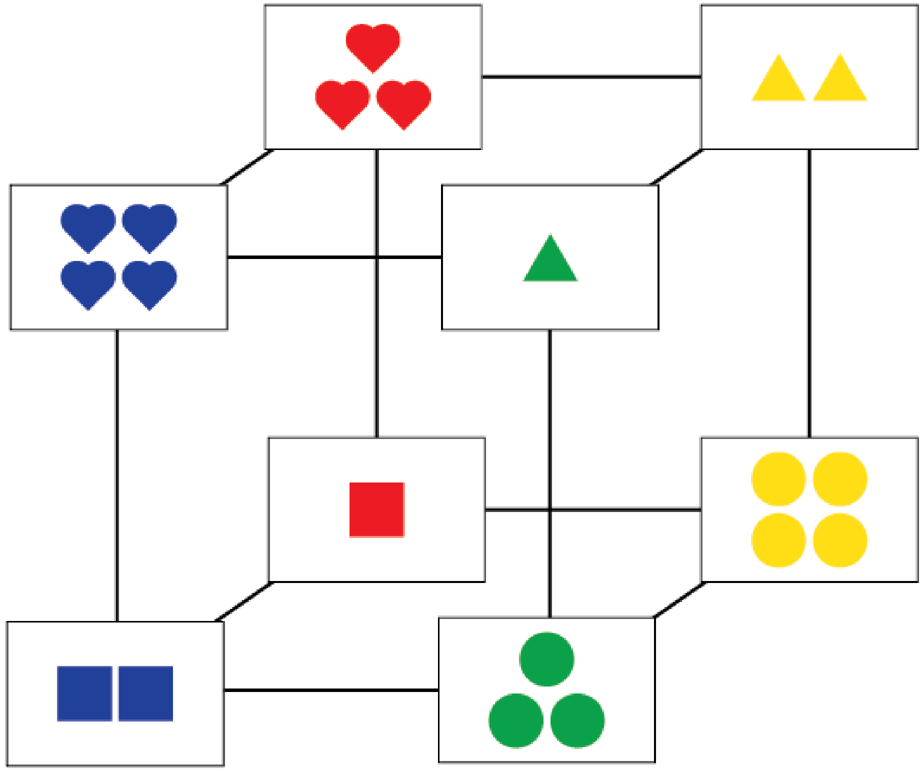
- *Proof:*
  - 3 cards = a triple
  - Either complete (4 per quad) or incomplete
  - At most $\binom{k}{3}$ complete triples, so at most $\frac{\binom{k}{3}}{4}$ quads



Powers of 2

Theorem: For an integer n such that $2 \leq n \leq d$, we have $\mathrm{q}(2^{\mathrm{n}}, \mathrm{d}) = \frac{\binom{2^n}{3}}{4}$.

- *Proof:*
  - Choose n linearly independent vectors (these exist since $n \leq d$)
  - Linear subspaces are closed under odd sums (counting 0 as an independent vector)
  - Therefore, every triple is complete



A cube of quads cards with 14 quads.

QuadFinder Code

k	d	$q(k,d)$
4	6	1
5	6	1
6	6	3
7	6	7
8	6	14
9	6	14
10	4	18

```
void iterate(int *a, int i, int size, int n, ll &count) {
    int start = 0;
    if (i > 0) {
        start = a[i-1]+1;
    }
    for (a[i] = start; a[i] < size; a[i]++) {
        if (i == n-1) {
            vector<int> subset;
            for (int k = 0; k < n; k++) {
                subset.push_back(a[k]);
            }
            ll curcount = 0;
            for (int i = 0; i < n; i++) {
                for (int j = i+1; j < n; j++) {
                    for (int k = j+1; k < n; k++) {
                        for (int l = k+1; l < n; l++) {
                            int cur = 0;
                            cur ^= subset[i];
                            cur ^= subset[j];
                            cur ^= subset[k];
                            cur ^= subset[l];
                            if (cur == 0) {
                                curcount++;
                            }
                            count = max(count, curcount);
                        }
                    }
                }
            }
        }
        else {
            iterate(a, i+1, size, n, count);
        }
    }
}
```

QuadFinder Iterate Function

Rohan’s Quads Code

- Code written in Rust, original written in C++
- Improved upon original QuadFinder code
  - QuadFinder: looks at every subset of k cards from a deck of 2^n cards
  - Rohan’s code: tries to find a subset of k cards from a deck of 2^n cards that has a target number of quads, if one exists; exits the loop if one is found and stores the list of cards
- Also has a function that finds multiple target quads counts
- Allows us to refine best counts instead of performing all computations at once
  - Requires running the code multiple times

Conjectures

I. For $k \leq 2^d$, we have $q(k,d) = q(k,d+1)$.

II. We have $q(2^n+1,d) = q(2^n,d)$.

III. We have $q(2^n-1,d) = q(2^n,d) - \frac{\binom{2^n-1}{2}}{3}$.

IV. We can add a card to a maximally packed set of k cards, to get a maximally packed set of $k+1$ cards.

Based on our results from our code for small k, Conjectures I, II, and III appear to hold.

Conjecture IV was proposed as this is how we were able to find results for $4 \leq k \leq 6$.

Acknowledgements

This work was possible thanks to the 2024 Polymath Jr REU, which is funded by the NSF.

Thank you to our Polymath Jr project mentors, Dr. Lauren Rose, Dr. Timothy E. Goldberg, and Daniel Rose-Levine for supporting us.

Thank you to my Polymath Jr colleagues Dani Catala, Marcus Chung, and Sarah Covey for their wonderful contributions.

Thank you to my colleague Rohan Ridenour for greatly improving the original QuadsFinder code.

Thank you to the organizers of this meeting and this session!

Links and References

Taiki Aiba. GitHub Repository for QuadsFinder. https://github.com/detoasty3/quadsfinder.

Julia Crager et al. How many cards should you lay out in a game of EvenQuads: A detailed study of caps in AG(n,2). https://arxiv.org/pdf/2212.05353.

Liz McMahon et al. The Joy of SET: The Many Mathematical Dimensions of a Seemingly Simple Card Game.

Rohan Ridenour. GitHub Repository for Quads. https://github.com/CatsAreFluffy/quads.

Lauren Rose. GitHub Repository for Quads. https://drlaurenrose.github.io/QUADS.

Daniel Rose-Levine. Results on k-Covers in the Game Quads.

Association for Women in Mathematics. EvenQuads Website. https://awm-math.org/publications/playing-cards.

Thank you!

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Any questions?